Lab Report - 8

TELEMETRY AND REMOTE MOTOR CONTROL USING CAN BUS

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1. Objective

The experiment develops a two-node telemetry and remote motor-control system using TM4C123GH6PM LaunchPads on EduARM4 trainer boards. Node 1 reads a potentiometer, a DHT11 temperature sensor, and SW1/SW2, then transmits their status over a 500 kbps CAN bus every 100 ms. Node 2 receives the telemetry and controls two DC motors through an L298N driver. Each board displays its local status on a 16x2 LCD. The programs, node1_sensor.c and node2_actuator.c, use register-level C with tm4c123gh6pm.h and the default 16 MHz clock.

2. Hardware & GPIO Configuration

Both boards use the internal CAN0 controller with MCP2551 transceivers: PE5 is CAN0 TX and PE4 is CAN0 RX. Each transceiver is powered from 5 V; its RX output reaches PE4 through 1 kohm, with 2.2 kohm from PE4 to ground. CANH and CANL connect between nodes, with 120 ohm termination at each end and common grounds. Node 1 uses PE3/AIN0 for the potentiometer, PD2 for DHT11 DATA, and PF4/PF0 for SW1/SW2. On Node 2, PF2/M1PWM6 and PF3/M1PWM7 drive L298N ENA and ENB; PA2/PA3 and PA4/PA5 drive the respective direction inputs. ENA/ENB jumpers are removed, and the motors use an external supply. Both LCDs use PA6/PA7 for RS/E and PB0-PB7 for data, with RW grounded and EduARM4 J5 shorted at 3-4.

3. Sensor Acquisition and CAN Telemetry

Node 1 converts the potentiometer's 12-bit ADC result to 0-100%. The DHT11 uses a digital single-wire interface and is read every 2 s. Debounced SW1 and SW2 presses toggle the stored Motor-1 and Motor-2 direction states. One standard 11-bit CAN frame, identifier 0x100 and DLC 5, is sent every 100 ms: byte 0 carries potentiometer percentage; bytes 1-2 carry temperature high and low bytes; byte 3 contains Motor-1 direction in bit 0, Motor-2 direction in bit 1, and sensor fault in bit 2; byte 4 is a rolling counter. Direction 0 denotes clockwise and 1 anticlockwise. At 16 MHz, prescaler 2, TSEG1 = 12, TSEG2 = 3, and SJW = 2 give CAN0_BIT_R = 0x2B41 and an approximately 81% sample point.

4. PWM Motor Control and Link Timeout

Node 2 decodes the received frame and generates independent 1 kHz PWM outputs using PWM1 Generator 3. Motor-1 duty cycle equals the received potentiometer percentage. Motor-2 remains off at or below 20 degrees Celsius; above 20 degrees Celsius, its duty command follows 30 + (T - 20) x 70/30 percent, capped at 100% at 50 degrees Celsius and above. Thus, 27 degrees Celsius gives approximately 46% duty. The received direction bits set the L298N input pairs, allowing SW1 and SW2 to reverse the two motors independently. Direction states are transmitted rather than repeated toggle commands. If no frame arrives for 500 ms, both motors stop and the LCD displays NO CAN DATA.

5. LCD Status, Diagnostics, and Operation

Node 1 shows temperature and potentiometer percentage on the first LCD row, for example T:027°C P:045%, and S1:CW S2:ACW on the second row. Node 2 shows M1:045% CW and M2:046% ACW on its two rows. DHT11 failures replace the temperature with E1-E5: no response, line-stuck errors, bit-timing failure, or checksum failure. The displayed motor percentages represent PWM commands. A separate CCS project is built and flashed for each board using the DEBUG power setting and debug USB port. With the CAN bus and motor supply connected, turning the potentiometer changes Motor-1 speed, warming the DHT11 changes Motor-2 speed, and SW1/SW2 change motor directions.

6. Conclusion

The system combines ADC acquisition, digital temperature sensing, native CAN communication, dual PWM motor control, and LCD feedback. Periodic telemetry carries sensor values, direction states, a fault flag, and a rolling counter in one compact frame. Independent motor commands and a 500 ms communication timeout support coordinated remote actuation, while LCD status and DHT11 error codes make operation and troubleshooting clear.